ZEYNEP TEKİN

Data Scientist | AI & ML Engineer | Data Analyst | Astronomy Data Research (Intro2Astro Alumni)

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LinkedIn • GitHub • Kaggle • Hugging Face

PROFESSIONAL SUMMARY

Early-career Data Scientist, ML Engineer, and AI Researcher with applied training in Python, statistics, machine learning, deep learning, transformers, large language models, MLOps, and project-based development. Completed hands-on data science training and the Intro2Astro 2026 program, with experience translating data into reproducible analyses and research-ready outputs. Actively designed, developed, and published nearly 60 open-source AI models and data-driven projects across GitHub, Hugging Face, and Kaggle. Passionate about building scalable intelligent systems, predictive models, and astronomy data analyses that achieve quantified results.Languages: Turkish (Native), English (Professional Working Proficiency - strong reading/technical comprehension)

CORE TECHNICAL SKILLS

Programming & Data	Python, Jupyter/Anaconda, Google Colab, Git/GitHub, data cleaning/munging, feature engineering, exploratory data analysis, visualization
Statistics & ML	Regression (linear, polynomial, ridge, lasso, ElasticNet), classification, clustering, anomaly detection, SVM, PCA, train/test split, accuracy, recall, confusion matrix
Deep Learning & AI	TensorFlow, ANN/DNN, CNN, RNN/LSTM,NLP, sentiment analysis, recommendation systems, computer vision/OpenCV, time series, graph theory
LLM / RAG / Agents	Transformers, self-attention, prompt engineering, embeddings, document chunking, vector retrieval, semantic search, retrieval-augmented generation, tool/function calling, agent workflows.
MLOps & Deployment	End-to-end ML pipelines,data governance/compliance, containers, cross-platform model deployment, version control
Specialized Topics	AutoML (PyCaret, AutoKeras, Dask-ML), reinforcement learning, GIS/spatial analysis, transfer learning
Astronomy Research	Python for astronomy, scientific plotting, exoplanet transit light curves, radial-velocity modeling, online archive queries (Gaia, MAST, NASA Exoplanet Archive), scientific paper reading

SELECTED PROJECTS

A collection of 50+ open-source ML, AI, and data-driven projects spanning NLP, Deep Learning, Time Series, Recommendation Systems, Predictive Modeling, Exploratory Data Analysis, and Scientific Computing. Featured highlights are detailed below; the complete portfolio is available on GitHub, Hugging Face, and Kaggle.

Bank Churn Dataset / Binary Classification (Machine Learning)

Developed an XGBClassifier model to analyze customer churn dynamics in the banking sector, achieving a benchmark-winning AUC score of 0.88970 and a weighted F1-score of 0.8565 across seven algorithms. Configured optimal parameters (learning_rate=0.03, n_estimators=1000, max_depth=6) with an early_stopping_rounds=100 mechanism to perfectly protect against overfitting. Utilized RobustScaler to neutralize outlier effects and mapped high-risk customer segments using engineered features like IsActiveWithCard.

LMSYS - Chatbot Arena Human Preference Predictions (LLM & Evaluation)

LMSYS - Chatbot Arena Human Preference Predictions (LLM & Evaluation)Developed a hybrid model to predict AI response quality, achieving a Log Loss score of 1.0154 that outperformed the baseline. Integrated statistical text features—such as length difference ratio (len_diff_ratio), word count, presence of code blocks, and punctuation density—with semantic TF-IDF data. Analyzed prompt-response alignment (common_ratio) and structured probability matrices to accurately reflect competitive uncertainty.

Predict Customer Churn (Machine Learning)

Executed an end-to-end churn prediction pipeline on a 600k-row dataset, where XGBoost emerged as the superior model with a robust ROC_AUC of 0.9144 and a Recall of 0.82. Integrated demographic (Gender, Partner, Dependents), behavioral, and engineered signals (Risk_Score with 0.51 correlation, AvgMonthly, IsNewCustomer). Benchmarked LightGBM and CatBoost to validate Gradient Boosting Decision Trees (GBDTs) as the optimal mathematical fit to capture complex customer churn signals.

H&M Personalized Fashion Recommendations (RAG & Retrieval Strategy)

Developed a large-scale personalized recommendation system by analyzing millions of transaction records within the H&M dataset. Transformed customer demographics (age_group) and price distributions into high-fidelity features, deploying a LightGBM algorithm optimized with joblib compression techniques for massive data volumes. Architected a multi-layered Hybrid recommendation strategy integrating individual loyalty history, model intelligence, and age-specific trends into final 12-item recommendation lists to serve as a robust context-retrieval pipeline.

Demand Forecasting & Inventory Optimization (Predictive AI & Supply Chain)

Conducted an Inventory Optimization Simulation using the 5-day forecasts generated by the model to transition from reactive stocking to a data-driven inventory policy. Calculated a Safety Stock of 38.11 units to act as a buffer against supply chain uncertainties, an automated Reorder Point at 157.85 units as a critical trigger to prevent stockouts, and an Order Quantity of 239 units to optimize holding costs and product availability.

Store Sales and Profit Analysis (Predictive AI & Finance)

Trained an XGBRegressor model to predict the profit variable, demonstrating high performance with an R^2 score of 0.791 by constraining outliers. Integrated strategic features such as 'Price Per Unit' and 'Total Discount Amount'. Conducted operational analysis confirming that a 20% discount rate serves as a critical break-even point for the business, while evaluating segment-based profit margins and regional discount thresholds.

TRAINING & CERTIFICATES

Data Science, Machine Learning & AI Bootcamp - Mastering Applied Data Science with Deep Learning (2026)

Certificate after completion

- Hands-on, project-based training spanning statistics, Python, data science, machine learning, deep learning, AI, AutoML, reinforcement learning, GIS, quantum ML, transformers/LLMs, MLOps, and capstone/project presentations.
- Data science foundations: Statistics 101; Python 101; EDA; data visualization and information analysis; data munging; feature engineering; end-to-end problem solving and model evaluation.
- Machine learning: supervised, unsupervised, and reinforcement learning; regression families; classification and clustering; Naive Bayes/Bernoulli/decision trees; anomaly detection; SVM; PCA; train/test splits and K-fold validation.
- AI/deep learning: NLP and sentiment analysis; Hadoop/Spark; recommendation systems; OpenCV face recognition; TensorFlow/time series; ANN/DNN/CNN/RNN/LSTM/GAN; 3D computer vision; social-network analysis/page ranking; graph theory.
- Advanced modules: AutoML (PyCaret, AutoKeras, H2O.ai, Dask-ML); GIS/spatial analysis; reinforcement-learning concepts and policy gradients; QML/Qiskit; transfer learning; transformers/self-attention; LLMs and prompt engineering; MLOps/CI-CD, governance, containers, Docker and Kubernetes.
- Applied environment included Jupyter/Anaconda, Google Colab, Visual Studio Code, GitHub, Hugging Face, IBM Qiskit, Apache Spark, PythonAnywhere, Kaggle, and cloud/ML platforms.

EXPERIENCE & RESEARCH

Intro2Astro 2026 - Introduction to Astronomy Research

Completed introductory astronomy research training designed to prepare participants tramp quickly into formal research projects and research internships.

Developed practical research skills in Python programming, scientific plotting with real astronomical data, exoplanet transit light-curve fitting, radial-velocity modeling, and querying astronomy archives such as Gaia, MAST, and the NASA Exoplanet Archive.

Practiced scientific paper reading, research discussion, short-form scientific writing, and research/career preparation for opportunities such as REUs and graduate study; explored citizen-science pathways.

EDUCATION

Computer Programming | Anadolu University — *2026* – Present

M.Sc. in Forensic Sciences | Hitit University — 2019 – 2022

B.A. in Sociology (Honor) | Ankara Yıldırım Beyazıt Univ. — 2013 – 2017

ATS KEYWORD BANK - USE SELECTIVELY

Python • pandas • NumPy • scikit-learn • TensorFlow • deep learning • NLP • transformers • LLM • prompt engineering • embeddings • vector search • semantic retrieval • AI agents • tool calling • evaluation • machine learning • feature engineering • model validation • MLOps •GitHub • Jupyter • Hugging Face • OpenCV • Spark • AutoML • GIS •astronomy data analysis • exoplanets • Gaia • MAST